Homework 2

Due Thursday, September 17, 2026 by 11:59 PM on Gradescope

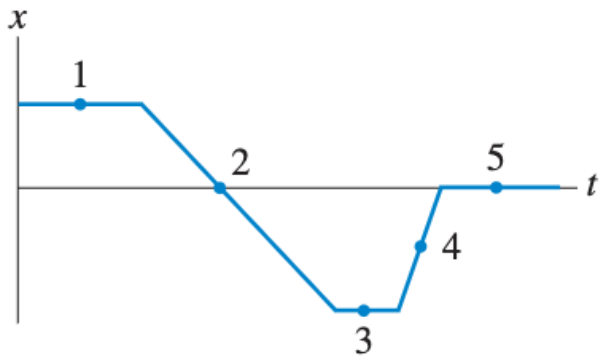
Instructions

- Include your full name and Coyote ID on the top right of the first page
- Clearly circle or box your answer for any question involving calculations. If the problem is multipart, circle or box answers for each part.
- Work should be neatly written and easy to follow.
- You must tag each question on Gradescope. If you do not tag your work for a given problem, you will not receive credit for it.

Problems

This homework contains problems related to 1D motion and 1D kinematics.

Figure 1



1. Figure 1 shows the position vs. time graph for a moving object. At which numbered point or points is the object:
   a. moving the slowest?
   b. moving the fastest?
   c. at rest?
   d. moving to the left?

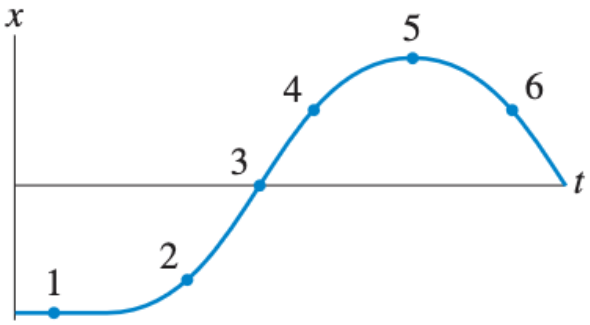
(a) 2. The object is still moving, but the magnitude of the slope of the position-versus-time curve is smaller than at 4.

(b) 4. The slope is greatest at 4.

(c) At points 1, 3, and 5 the slope of the curve is zero, so the object is not moving.

(d) At point B the slope is negative, so the object is moving to the left.

Figure 2



2. Figure 2 shows the position vs. time graph for a moving object. At which numbered point or points is the object:
   a. moving the slowest?
   b. moving the fastest?
   c. at rest?
   d. moving to the left?

(a) The slope of the position-versus-time graph is greatest at 3, so the object is moving fastest at this point.

(b) The slope is negative at point 6, meaning the object is moving to the left there.

(c) At point 6 the slope is increasing in magnitude (getting more negative), meaning that the object is speeding up to the left.

(d) At point 5 the object is not moving since the slope is zero. Before point 5, the slope is positive, while after 5 it is negative, so the object is turning around at 5.

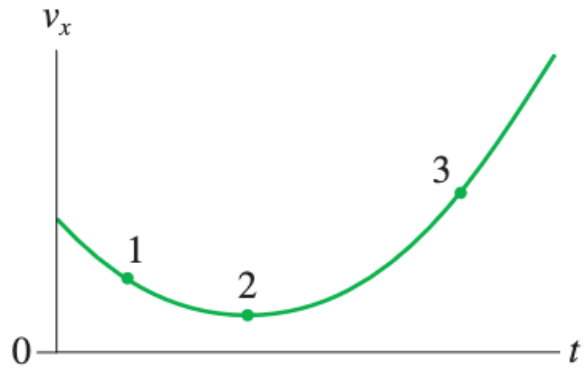
3. You're driving along the highway at a steady speed of $60\ \mathrm{mi/h}$ when another driver decides to pass you. At the moment when the front of his car is exactly even with the front of your car, and you turn your head to smile at him, do the two cars have equal velocity? Explain.

No, though your position is the same your velocities are different (otherwise they wouldn't pass you). Think about as though you are driving the car, you look down and you see the speedometer read $60\ \mathrm{mi/h}$, someone is passing you, do you think they are driving faster or slower than you? Faster because they are passing you. They look down at their speedometer and it should be more than $60\ \mathrm{mi/h}$. Recall the speedometer reads your instantaneous speed.

4. A bicycle is traveling east. Can its acceleration vector ever point west? Explain.

Yes. The acceleration vector will point west when the bicycle is slowing down while traveling east.

Figure 3



5. Figure 3 shows the velocity-vs.-time graph for a moving object. At which number point or points is the object:
   a. speeding up?
   b. slowing down?
   c. moving to the left?
   d. moving to the right?

(a) The vertical axis of the graph is velocity, not position. The object is speeding up where the velocity is increasing; this is the case at point 3.

(b) The object is slowing down at point 1 because the velocity in the x direction is getting smaller.

(c) The graph of velocity is always above the t axis, so the velocity is always positive, or in the direction to the right. At none of the points 1, 2, or 3 is it moving to the left.

(d) The object is moving to the right at all three points because the velocity is positive at all three points.

6.

a. What constant acceleration, in SI units, must a car have to go from zero to $60\ \text{mi/h}$ in $10\ \text{s}$?

b. How far has the care traveled when it reaches $60\ \text{mi/h}$? Give your answer both in SI units and in feet.

MODEL: Represent the car as a particle.

SOLVE: **(a)** First, we will convert units:

$$60\frac{\text{mi}}{\text{h}}\times\frac{1\ \text{h}}{3600\ \text{s}}\times\frac{1610\ \text{m}}{1\ \text{mi}}=27\ \text{m/s}$$

The motion is constant acceleration, so

$$v_{1x}=v_{0x}+a_x\Delta t\Rightarrow a_x=\frac{v_{1x}-v_{0x}}{\Delta t}=\frac{(27\ \text{m/s}-0\ \text{m/s})}{10\ \text{s}}=2.7\ \text{m/s}^2$$

(b) The distance is calculated as follows:

$$x_1=x_0+v_{0x}\Delta t+\frac{1}{2}a_x(\Delta t)^2=\frac{1}{2}a_x(\Delta t)^2=1.3\times10^2\ \text{m}=4.3\times10^2\ \text{ft}$$

REVIEW: We expect a car driving for 10 s to move many feet. We would expect an answer in the hundreds. Our answer seems reasonable.

7. It has been proposed that a very small probe could be sent to a nearby star system by using a powerful leaser beam, fired from an earth-orbiting satellite, to push on a lightweight “solar sail.” Very high speeds could be reached in the vacuum of space by a fairly modest acceleration that continues for a long interval of time.
   a. Write an expression for the constant acceleration a_x an object needs to reach velocity $v_{\max}$ in time t_{push}, starting from rest.
   b. Write an expression in terms of $v_{\max}$ and t_{push} for the distance d the object travels during this time.
   c. For the mission to be feasible, the probe needs to reach 10% of the speed of light after being pushed for $1.0\ \text{year}$. The probe would then coast the rest of the way. What constant acceleration is needed? (Note the speed of light and other useful data needed to solve problems are given inside the root and back covers of the book)
   d. What fraction of a light year will the probe have traveled at the end of the year? A light year (ly) is the distance traveled by light in $1\ \text{year}$

MODEL: Throughout the problem, we will ignore the spatial dimensions of the probe, and treat it like a particle.

VISUALIZE: We will use $v_x = v_{0x} + a_x \Delta t$ as well as $v_x^2 = v_{0x}^2 + 2a_x \Delta x$.

SOLVE: **(a)** Starting from rest, we have $a_x = v_{\max} / t_{\text{push}}$.

(b) We know $v_x^2 = v_{0x}^2 + 2a_x \Delta x$. Combining this with the result from (a), we have

$$v_{\max}^2 = v_{0x}^2 + 2a_x \Delta x = 0 + 2\left(\frac{v_{\max}}{t_{\text{push}}}\right)\Delta x \Rightarrow \Delta x = \frac{1}{2} v_{\max} t_{\text{push}}$$

(c) The speed of light is 3.00×10^8 m/s, so the acceleration required is

$$a_x = \frac{(0.10)c}{t_{\text{push}}} = \frac{(0.10)\left(3.00\times10^8 \text{ m/s}\right)}{(1.0 \text{ y})(365 \text{ d/y})(24 \text{ h/d})(3600 \text{ s/h})} = 0.95 \text{ m/s}^2.$$

(d) Using the result from part (b), we have

$$\frac{\Delta x}{(1c \cdot \text{y})} = \frac{1}{2(1c \cdot \text{y})} v_{\max} t_{\text{push}} = \frac{1}{2}\frac{(0.10)c}{(1c \cdot \text{y})}(1.0 \text{ y}) = \frac{0.10}{2} = 0.050$$

Or 5.0% of a light year.

REVIEW: Since the probe never travels faster than 1/10 of the speed of light, we expect the distance it crosses in a year to be much less than a light-year. Our answer, 5.0% of a light year, is reasonable.

8. When you sneeze, the air in your lungs accelerates from rest to $150\,\text{km/h}$ in approximately $0.50\,\text{s}$. What is the magnitude of the acceleration of the air in m/s^2?

MODEL: Model the air as a particle.

VISUALIZE: Use the definition of acceleration and then convert units.

SOLVE:

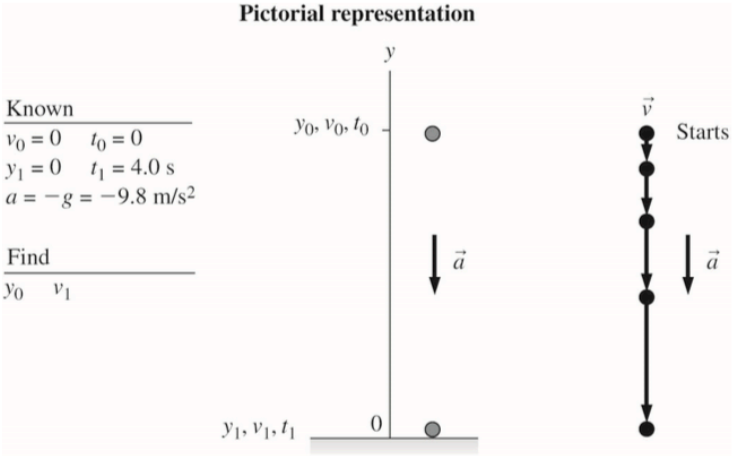
$$a_x = \frac{\Delta v_x}{\Delta t} = \frac{150\text{ km/h}}{0.50\text{ s}}\left(\frac{1000\text{ m}}{1\text{ km}}\right)\left(\frac{1\text{ h}}{60\text{ min}}\right)\left(\frac{1\text{ min}}{60\text{ s}}\right) = 83\text{ m/s}^2$$

REVIEW: 83 m/s^2 is a remarkable acceleration.

9. Ball bearings are made by letting spherical drops of molten metal fall inside a tall tower—called a *shot tower*—and solidify as they fall.
   a. If a bearing needs $4.0\ \text{s}$ to solidify enough for impact, how high must the tower be?
   b. What is the bearing's impact velocity?

MODEL: Represent the spherical drop of molten metal as a particle.

VISUALIZE:



SOLVE: **(a)** The shot is in free fall, so we can use free-fall kinematics with $a = -g$. The height must be such that the shot takes 4.0 s to fall, so we choose $t_1 = 4.0$ s. Then,

$$y_1 = y_0 + v_0(t_1 - t_0) - \frac{1}{2}g(t_1 - t_0)^2 \Rightarrow y_0 = \frac{1}{2}gt_1^2 = \frac{1}{2}(9.8\ \text{m/s}^2)(4.0\ \text{s})^2 = 78.4\ \text{m}$$

(b) The impact velocity is $v_1 = v_0 - g(t_1 - t_0) = -gt_1 = -39.2$ m/s.

REVIEW: Note the minus sign. The question asked for *velocity*, not speed, and the y-component of $\vec{v}$ is negative because the vector points downward.

10. A rock is tossed straight up from ground level with a speed of $20\,\mathrm{m/s}$. When it returns, it falls into a hole $10\,\mathrm{m}$ deep.

a. What is the rock's speed as it hits the bottom of the hole?

Picture tossing a rock straight up. It rises, slows, stops for an instant at the top, then falls back down and lands in your hand. When it returns to the initial position, the magnitude of the velocity is the same as when it was tossed but pointing in the opposite direction. The reason for this is because the only influence on the rock is gravity which slows the rock at the same rate that it speeds it up on the return trip. So, if the rock goes the same distance both directions, it's change in speed is necessarily the same. So to find the final velocity we can use $v_\mathrm{f}^2 = v_\mathrm{i}^2 + 2a\Delta x$. We are trying to find v_f, we know that $v_\mathrm{i} = -20\,\mathrm{m/s}$, $a = -g$, and $\Delta x = -10\,\mathrm{m} - 0\,\mathrm{m} = -10\,\mathrm{m}$, solving for $v_f = \pm\sqrt{(-20\,\mathrm{m/s})^2 + 2 \cdot 9.80 \cdot 10\,\mathrm{m^2/s^2}} = \pm 24\,\mathrm{m/s}$. We choose the sign based on the problem, since the rock was moving in the $-y$ direction the velocity must also be negative.

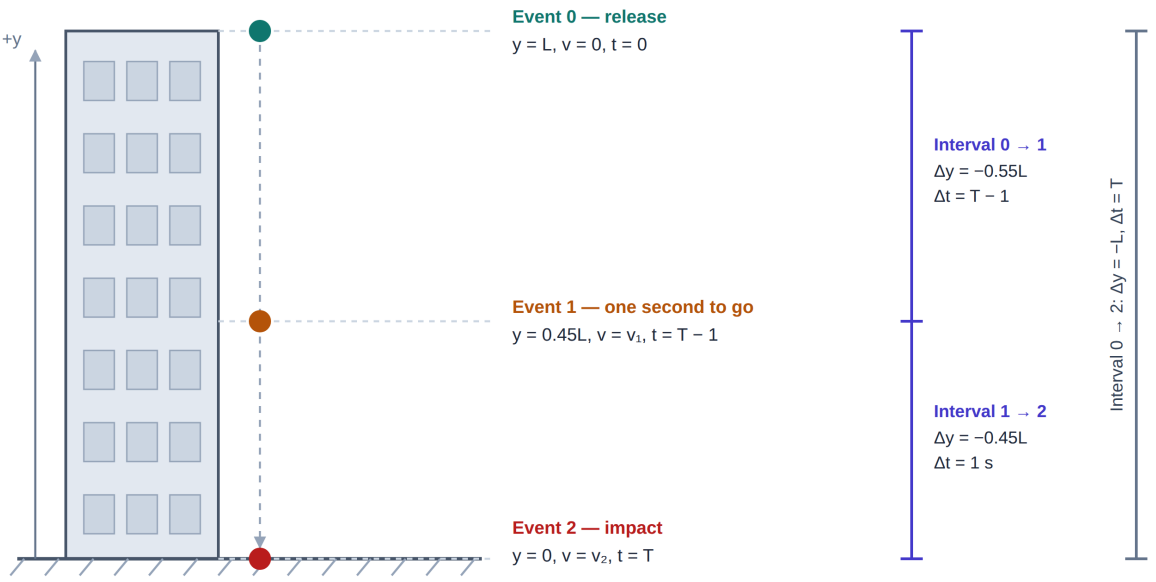
b. How long is the rock in the air, from the instant it is released until it hits the bottom of the hole?

To find the time we have three intervals, the first is from when the rock is tossed to where it stops in the air, the second is where it falls back down to ground level, and the third is from ground level to the bottom of the hole. Using the same argument as before, the time for the first two intervals must be the same. So we find the velocity for one of the intervals, multiple it by two, then find the time for the last interval and add both times together for the total time. Since we know the initial and final velocity

as well as the acceleration we will use $v_\mathrm{f} = v_\mathrm{i} + a\Delta t$.

$\Delta t = \dfrac{v_\mathrm{f} - v_\mathrm{i}}{a} = \dfrac{20\,\mathrm{m/s}}{9.80\,\mathrm{m/s^2}} = 2.04\,\mathrm{s}$. For the hole we can use the same formula with the final velocity we just computed. This gives $0.450\,\mathrm{s}$. So the total time is $2 \cdot 2.04\,\mathrm{s} + 0.450\,\mathrm{s} = 4.5\,\mathrm{s}$

11. A rock is dropped from the top of a tall building. The rock's displacement in the last second before it hits the ground is 45% of the entire distance it falls. How tall is the building?



See the "Organizing a kinematics problem - events and intervals" for a full discussion of this problem. Take up as positive.

- Full fall: $v_2^2 = v_0^2 + 2(-g)(-L) = 2gL$
- Last second: $v_2^2 = v_1^2 + 2(-g)(-0.45L) = v_1^2 + 2g(0.45L)$
- Eliminate g and L: Solve both equations for $2g$ and set equal $\frac{v_2^2}{L} = \frac{v_2^2 - v_1^2}{0.45L}$ then rearrange so the L's are on one side and the v's on

the other and simplify $0.45 = 1 - \frac{v_1^2}{v_2^2}$, finally solve for v_1 to get

$$v_1 = \sqrt{0.55}\, v_2$$

- Plug v_1 into $v_2 = v_1 + a\Delta$ for the one-second interval to find the result

  $$v_2 = -37.9\,\mathrm{m/s}$$

- Finally use v_2 in $v_2^2 = v_0^2 + 2a\Delta y$ to get $L = 73.4\,\mathrm{m}$

12. A car traveling at $30\ \text{m/s}$ runs out of gas while traveling up a 10° slope. How far up the hill will it coast before starting to roll back down?

MODEL: Represent the car as a particle.

VISUALIZE:

Known

$\theta = 10°$ $x_0 = 0$

$v_0 = 30$ m/s $t_0 = 0$

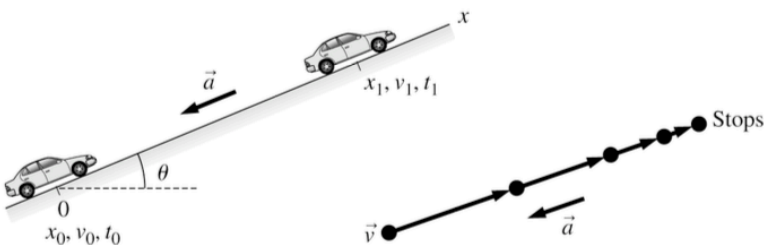
$v_1 = 0$ m/s

$a = -g\sin\theta = -1.7$ m/s^2

Find

x_1

Pictorial representation



SOLVE: Note that the problem "ends" at a turning point, where the car has an instantaneous speed of 0 m/s before rolling back down. The rolling back motion is *not* part of this problem. If we assume the car rolls without friction, then we have motion on a frictionless inclined plane with an acceleration $a = -g\sin\theta = -g\sin 10° = -1.7\ \text{m/s}^2$. Constant-acceleration kinematics gives

$$v_1^2 = v_0^2 + 2a(x_1 - x_0) \Rightarrow 0\ \text{m}^2/\text{s}^2 = v_0^2 + 2ax_1 \Rightarrow x_1 = -\frac{v_0^2}{2a} = -\frac{(30\ \text{m/s})^2}{2(-1.7\ \text{m/s}^2)} = 265\ \text{m}$$

Notice how the two negatives canceled to give a positive value for x_1.

REVIEW: We must include the minus sign because the $\vec{a}$ vector points *down* the slope, which is in the negative x-direction.

13. A bicycle coasting at $8.0\ \text{m/s}$ comes to a $5.0\ \text{m}$ long, $1.0\ \text{m}$ high ramp. What is the bicycle's speed as it leaves the top of the ramp?

MODEL: We will model the bicycle as a particle, ignoring its length and height. We will ignore rolling friction. We know an object free to slide on an incline experiences an acceleration of $a_x = -g\sin(\theta)$, where x is assumed to point up the incline.

VISUALIZE: Note that the angle between the incline and the ground is $\theta = \sin^{-1}(1/5) = 11.5°$, and the length is given as 5.0 m.

SOLVE: We can use $v_x^2 = v_{0x}^2 + 2a_x\Delta x$, inserting the acceleration from above to obtain

$$v_x = \sqrt{v_{0x}^2 + 2a_x\Delta x} = \sqrt{(8.0\ \text{m/s})^2 + 2(-9.8\ \text{m/s}^2)(\sin(\theta))(5.0\,\text{m})} = 6.7\ \text{m/s}$$

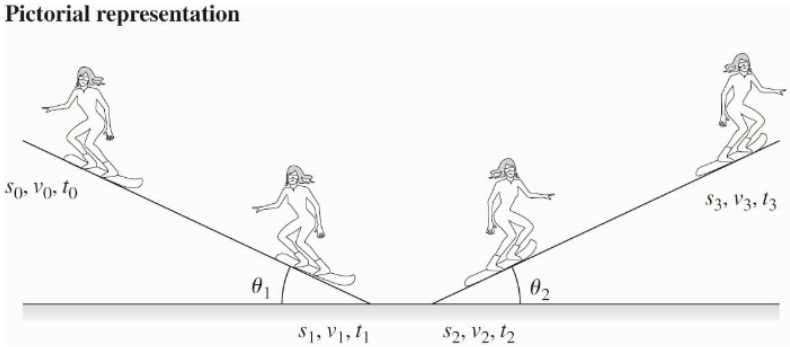
REVIEW: We expect the bicycle to slow down, so it is sensible that the answer we obtained is less than the original 8.0 m/s.

14. A snowboarder glides down a $50\ \text{m}$ long, 15° hill. She then glides horizontally for $10\ \text{m}$ before reaching a 25° upward slope. Assume the show is frictionless.
    a. What is her speed at the bottom of the hill?
    b. How far can she travel up the 25° slope?

MODEL: The snowboarder is a particle moving under constant-acceleration kinematic equations. The speed does not diminish on the horizontal section.

VISUALIZE: Note that our s-axis is positioned along both inclines.

Pictorial representation



Known
$s_0 = 0 \quad v_0 = 0 \quad t_0 = 0$
$\theta_1 = 15° \quad \theta_2 = 25°$
$v_2 = v_1$
$a = g \sin\theta$
y_1, v_1, t_1
y_2, v_2, t_2
Find
$s_3 - s_2$

SOLVE: **(a)** Using the following kinematic equation for the downward slope

$$v_1^2 = v_0^2 + 2a_0(s_1 - s_0) = (0\ \text{m/s})^2 + 2(9.8\ \text{m/s}^2)(\sin 15^\circ)(50\ \text{m} - 0\ \text{m}) \Rightarrow v_1 = 15.93\ \text{m/s}$$

which is reported as 16 m/s to two significant figures.

(b) Use the same equation again on the upward slope (with $v_2 = v_1$).

$$v_3^2 = v_2^2 + 2a_2(s_3 - s_2) = (0\ \text{m/s})^2 \Rightarrow$$

$$(s_3 - s_2) = \frac{-v_1^2}{2a_2} = \frac{-(15.93\ \text{m/s})^2}{2(-9.8\ \text{m/s}^2)(\sin 25^\circ)} = 31\ \text{m}$$

REVIEW: Because the upward slope is steeper we did not expect the snowboarder to travel as far up the slope.

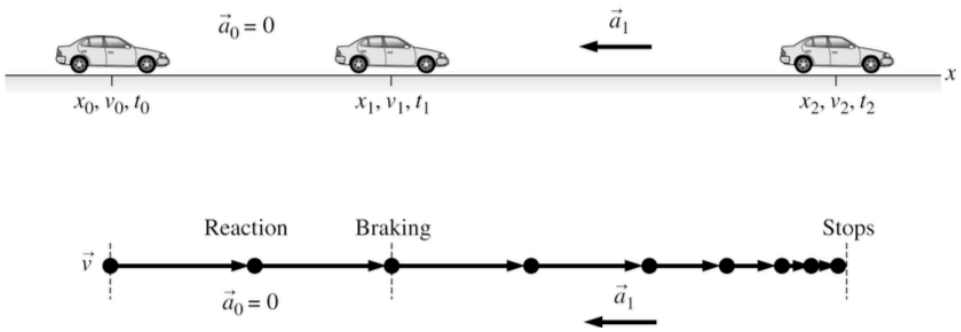
15. You're driving down the highway late one night at $20\ \mathrm{m/s}$ when a deer steps onto the road $35\ \mathrm{m}$ in front of you. Your reaction time before stepping on the breaks is $0.50\ \mathrm{s}$, and the maximum deceleration of your car is $10\ \mathrm{m/s^2}$.
    a. How much distance is between you and the deer when you come to a stop?
    b. What is the maximum speed you could have before stopping and still not hit the deer?

VISUALIZE:

Known
$x_0 = 0 \quad v_0 = 20\ \mathrm{m/s}$
$t_0 = 0\ \mathrm{s} \quad v_1 = 20\ \mathrm{m/s}$
$t_1 = 0.50\ \mathrm{s} \quad v_2 = 0$
$x_2 = 110\ \mathrm{m}$

Find
$x_1 \quad a_1 \quad t_2$

Pictorial representation



SOLVE: This is a two-part problem. During the reaction time,

$$x_1 = x_0 + v_0(t_1 - t_0) + 1/2 a_0(t_1 - t_0)^2$$
$$= 0\ \mathrm{m} + (20\ \mathrm{m/s})(0.50\ \mathrm{s} - 0\ \mathrm{s}) + 0\ \mathrm{m} = 10\ \mathrm{m}$$

After reacting, $x_2 - x_1 = 110\ \mathrm{m} - 10\ \mathrm{m} = 100\ \mathrm{m}$, that is, you are 100 m away from the intersection.

To stop successfully,

$$v_2^2 = v_1^2 + 2a_1(x_2 - x_1) \Rightarrow (0\ \mathrm{m/s})^2 = (20\ \mathrm{m/s})^2 + 2a_1(100\ \mathrm{m}) \Rightarrow a_1 = -2\ \mathrm{m/s^2}$$

16. The Boring Company has proposed transporting people between Los Angeles and San Francisco, a distance of $540\,\text{km}$, through an underground tube from which the air has been removed to eliminate drag. Small pods carrying four passengers would accelerate at $2.5\,\text{m/s}^2$ until reaching a speed of $330\,\text{m/s}$. Later they would break at $1.5\,\text{m/s}^2$. A launch of one pod per minute would transport 240 passengers per hour, roughly equivalent to one get plane per hour.

    a. What would be the trip time in minutes from Los Angeles to San Francisco?

Event	x	v	a	t
Launch	$0\,\text{m}$	$0\,\text{m/s}$	$2.5\,\text{m/s}^2$	$0\,\text{s}$
Cruising	x_1	$330\,\text{m/s}$	$0\,\text{m/s}^2$	t_1
Decelerating	x_2	$330\,\text{m/s}$	$-1.5\,\text{m/s}^2$	t_2
End	$540\times10^3\,\text{m}$	$0\,\text{m/s}$	$0\,\text{m/s}^2$	t_3

	Lauch —> Cruising	Cruising —> Decelerating	Decelerating —> End
Δx	x_1	$x_2 - x_1$	$540\times10^3\,\text{m} - x_2$
v_i	$0\,\text{m/s}$	$330\,\text{m/s}$	$330\,\text{m/s}$
v_f	$330\,\text{m/s}$	$330\,\text{m/s}$	$0\,\text{m/s}$
Δt	t_1	$t_2 - t_1$	$t_3 - t_2$

For launch —> cruising and decelerating —> end we can find the time using $v_\text{f} = v_\text{i} + a\Delta t$:

$$t_1 = \frac{330\,\text{m/s}}{2.5\,\text{m/s}^2} = 132\,\text{s},\ \Delta t_{23} = \frac{330\,\text{m/s}}{1.5\,\text{m/s}^2} = 220\,\text{s}$$

We also need to determine the displacements Δx_1 and Δx_{23} which we can compute from $v_f^2 = v_i^2 + 2a\Delta x$:

$$\Delta x_1 = \frac{(330\,\mathrm{m/s})^2}{5\,\mathrm{m/s^2}} = 21.780 \times 10^3\,\mathrm{m} = 21.78\,\mathrm{km}$$

$$\Delta x_{23} = \frac{(330\,\mathrm{m/s})^2}{3\,\mathrm{m/s^2}} = 36.30 \times 10^3\,\mathrm{m} = 36.30\,\mathrm{km}$$

For cruising —> decelerating we can find the time using the average velocity formula $v_{\mathrm{avg}} = \dfrac{\Delta x}{\Delta t}$, but we first need to find Δx by adding the displacements from the two accelerating sections and subtracting that from the total displacement

$$\Delta x_{03} - \Delta x_1 - \Delta x_{23} = 540 \times\ \mathrm{km} - 21.78\,\mathrm{km} - 36.30\,\mathrm{km} = 481.92\,\mathrm{km}$$

$$\Delta t_{12} = \frac{\Delta x}{v_{\mathrm{avg}}} = 1.46 \times 10^3\,\mathrm{s}$$

$$\Delta t_{03} = T = \Delta t_{01} + \Delta t_{12} + \Delta t_{23} = 1812\,\mathrm{s} = 30.2\,\mathrm{min}$$

b. How far apart would two adjacent pods be on the constant-speed segment of the journey?

Both pods accelerate at the same rate and the both stop accelerating at the same position, the only difference is that the first pod is moving at its constant velocity of $330\,\mathrm{m/s}$ for $1\,\mathrm{min}$ as the second pod is accelerating. After that they are both moving at the same speed and so their separation does not change. So the separation distance is

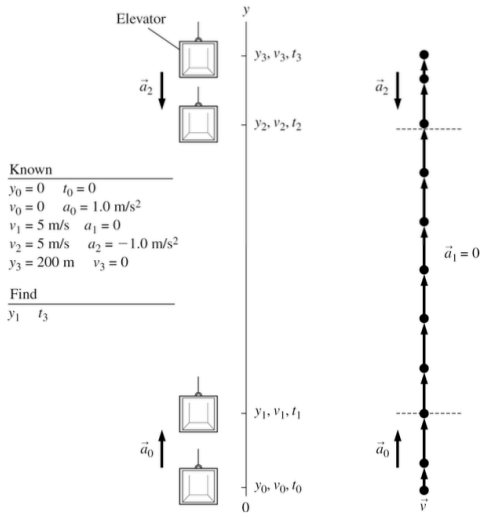
$$330\,\mathrm{m/s} \cdot 60\,\mathrm{s} = 19.8 \times 10^3\,\mathrm{m} = 19.8\,\mathrm{km}.$$

17. A hotel elevator ascends $200\ \text{m}$ with a maximum speed of $5.0\ \text{m/s}$. Its acceleration and deceleration both have a magnitude of $1.0\ \text{m/s}$.
    a. How far does the elevator move while accelerating to full speed from rest?
    b. How long does it take to make the complete trip from bottom to top?

MODEL: The elevator is a particle moving under constant-acceleration kinematic equations.

VISUALIZE:

Pictorial representation



SOLVE: **(a)** To calculate the distance to accelerate up:

$$(v_1)^2 = v_0^2 + 2a_0(y_0 - y_0) \Rightarrow (5\ \text{m/s})^2 = (0\ \text{m/s})^2 + 2(1\ \text{m/s}^2)(y_1 - 0\ \text{m}) \Rightarrow y_1 = 12.5\ \text{m}$$

(b) To calculate the time to accelerate up:

$$v_1 = v_0 + a_0(t_1 - t_0) \Rightarrow 5\ \text{m/s} = 0\ \text{m/s} + (1\ \text{m/s}^2)(t_1 - 0\ \text{s}) \Rightarrow t_1 = 5\ \text{s}$$

To calculate the distance to decelerate at the top:

$$v_3^2 = v_2^2 + 2a_2(y_3 - y_2) \Rightarrow (0\ \text{m/s})^2 = (5\ \text{m/s})^2 + 2(-1\ \text{m/s}^2)(y_3 - y_2) \Rightarrow y_3 - y_2 = 12.5\ \text{m}$$

To calculate the time to decelerate at the top:

$$v_3 = v_2 + a_2(t_3 - t_2) \Rightarrow 0\ \text{m/s} = 5\ \text{m/s} + (-1\ \text{m/s}^2)(t_3 - t_2) \Rightarrow t_3 - t_2 = 5\ \text{s}$$

The distance moved up at 5 m/s is

$$y_2 - y_1 = (y_3 - y_0) - (y_3 - y_2) - (y_1 - y_0) = 200\ \text{m} - 12.5\ \text{m} - 12.5\ \text{m} = 175\ \text{m}$$

The time to move up 175 m is given by

$$y_2 - y_1 = v_1(t_2 - t_1) + \frac{1}{2}a_1(t_2 - t_1)^2 \Rightarrow 175\ \text{m} = (5\ \text{m/s})(t_2 - t_1) \Rightarrow (t_2 - t_1) = 35\ \text{s}$$

To total time to move to the top is

$$(t_1 - t_0) + (t_2 - t_1) + (t_3 - t_2) = 5\ \text{s} + 35\ \text{s} + 5\ \text{s} = 45\ \text{s}$$

REVIEW: To cover a distance of 200 m at 5 m/s (ignoring acceleration and deceleration times) will require a time of 40 s. This is comparable to the time of 45 s for the entire trip as obtained above.